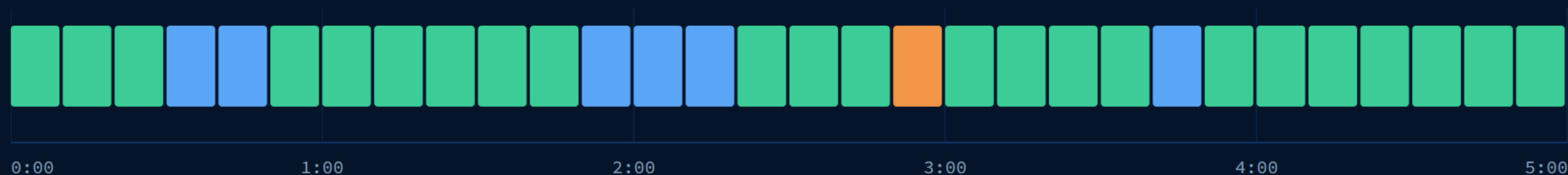


Pipeline

Your Vision ~~Model~~ Is Biased

Lessons from shipping real-time video detection

Serban Petrescu • Trilogy / TimeBack • Compass AI 2026



No teacher in the room, and no one watching the test

WHAT WE BUILD

- Trilogy's new products division: the software behind **Alpha School**
- A private US school: digital-first, AI-first, physical and online campuses
- No teachers: adaptive learning apps, and guides in the room
- Progress checked with periodic standardized tests

THE PROBLEM

- Students alone with devices **cheat**, idle, and work on the wrong thing
- Official third-party tests kept bringing surprises
- The answer: proctoring software

WE DETECT

- Cheating
- Wasted time
- Self-harm behaviors, raised to a safety officer

One billion students. Zero humans.

- The founder's goal: improve education for **1 billion students** in 20 years.
- Humans bootstrap the detectors. Every human left in the loop is one we are working to remove.
- Today, no human reviews a production detection before the student sees it.

1

engineer built it

2 mo

to real student traffic

140k

hours of video a month

3,400

students, last 30 days

335B

tokens a month

Eighteen months of mistakes. Every one was a bias we put there ourselves.

Three ways students cheat

UNAUTHORIZED DEVICE USE



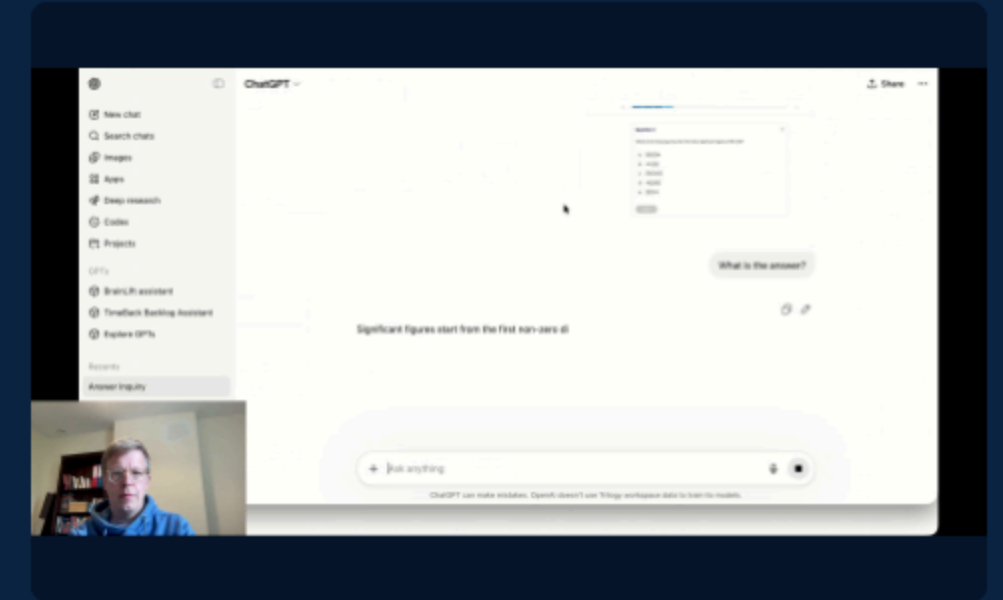
A handheld device, visibly in use. Judged from the webcam alone.

HELP FROM ANOTHER PERSON

“The answer is B.”

Another person supplies the answer to the active question.

UNAUTHORIZED APP USE



An external tool used to obtain the answer to the question on screen.

Every flagged behavior has an allowed twin

FLAGGED

Device use

Phone in hand

Help from another person

“The answer is B.”

App use

Pastes the question into ChatGPT, submits the answer it gives

ALLOWED

Phone lying on the desk, untouched

“Your screen froze, let me fix it.”

A calculator, on a math test that allows one

To the camera, each pair looks nearly identical.

Boundary rules from the production annotation instructions; the same rules humans are graded against.

Precision below 0.90 disqualifies a detector

0.90

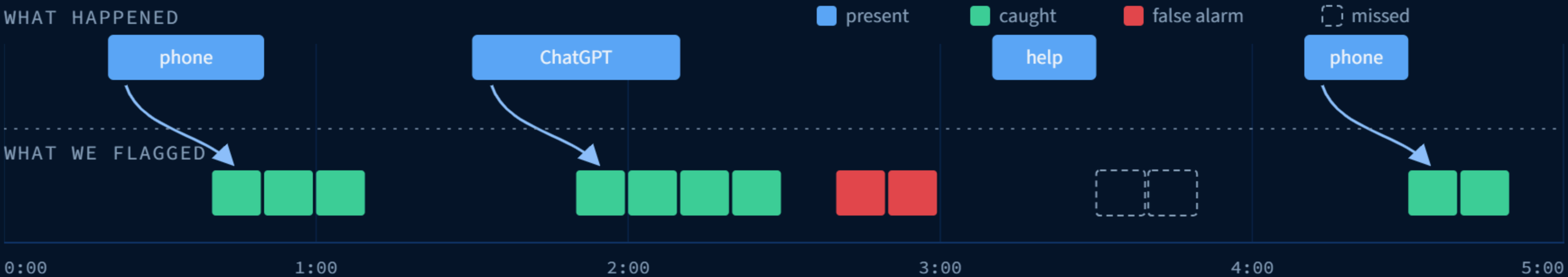
precision: of what we flag, the share that is real.
0.95 is the goal. The hard gate.

0.90

recall: of what happened, the share we catch.
Second priority.

≤2 min

latency, event to warning. Constant, never
accumulating.



No single flag ever punishes a student on its own.

We build gold ten seconds at a time

- Every week brought a new model and a new idea. Every demo looked good.
- Picking needs a harness: run every idea against the same sessions, scored the same way.
- Pre-trained or not: a quality claim needs a figure behind it.

FIVE REAL MINUTES OF GOLD: WORK ALONE, ONE SESSION



every block carries polarity, difficulty and a written reason. One of the orange blocks:

“The student observes something off-camera and speaks, maybe to herself. Surely about the test question though.”

We annotate in 10-second blocks to keep it simple. In around 13% of them, humans could not decide on the first pass.

Six annotators, one session, three verdicts on the same ten seconds



The same ten seconds. Opposite verdicts, all with confident reasons:

SERBAN **FLAGGED**

“Done fully by the peer. No input from the student.”

KSHAMA **FLAGGED**

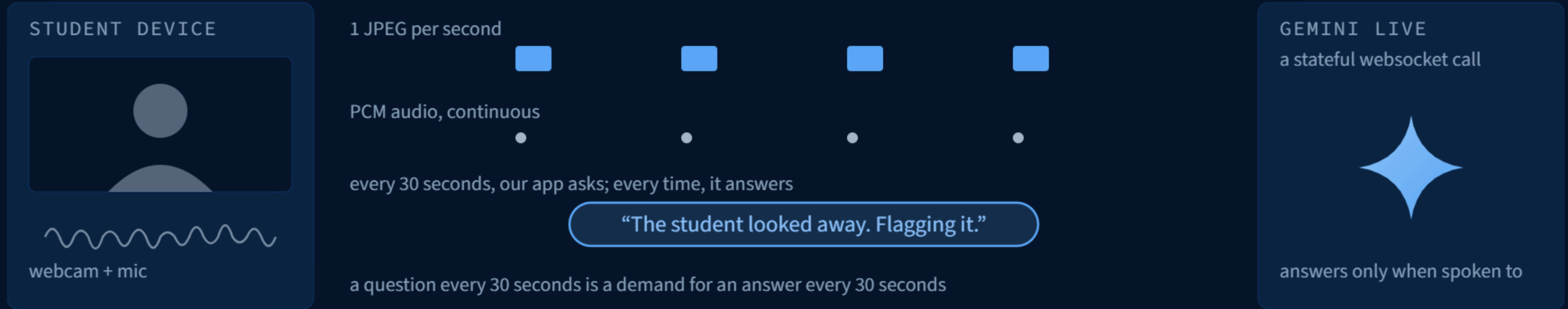
“An older sibling sits next to the student for the entire test.”

DULE **CLEAN**

“Three children are present, but nothing contributes to an answer.”

Disagreement was the point: every divergence told us which rule to write.

Attempt 1: a model that only speaks when spoken to



n/a

precision: never measured.

n/a

recall: never measured.

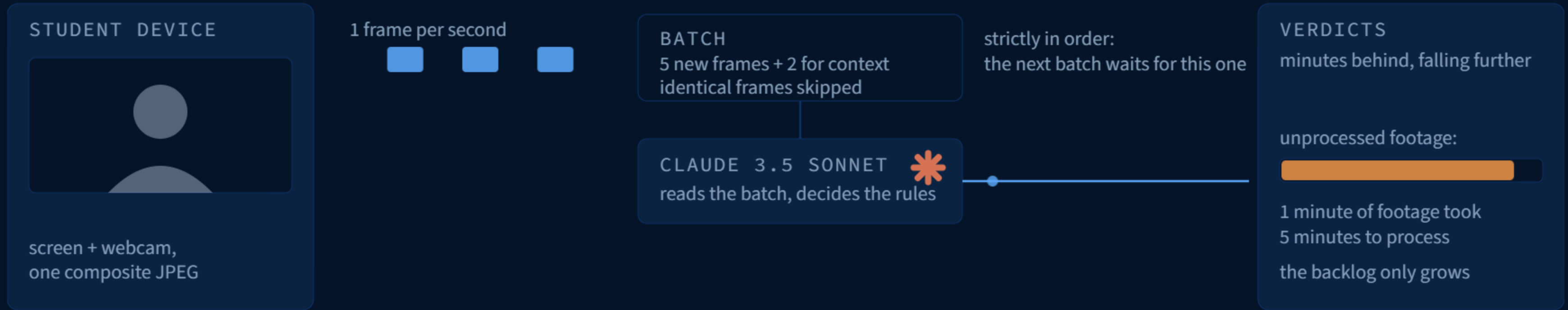
30 s

latency: the nudge cadence, by construction.

Ask a model every 30 seconds whether anything happened, and it will tell you something did.

Checked September 2026: Live sessions now run unlimited, the model may decline to answer, and input is still one JPEG per second. Replaying a live stream against gold is still impractical.

Attempt 2: stills gave 0.20 recall and a spiralling backlog



0.70

precision: far below the 0.90 gate.

0.20

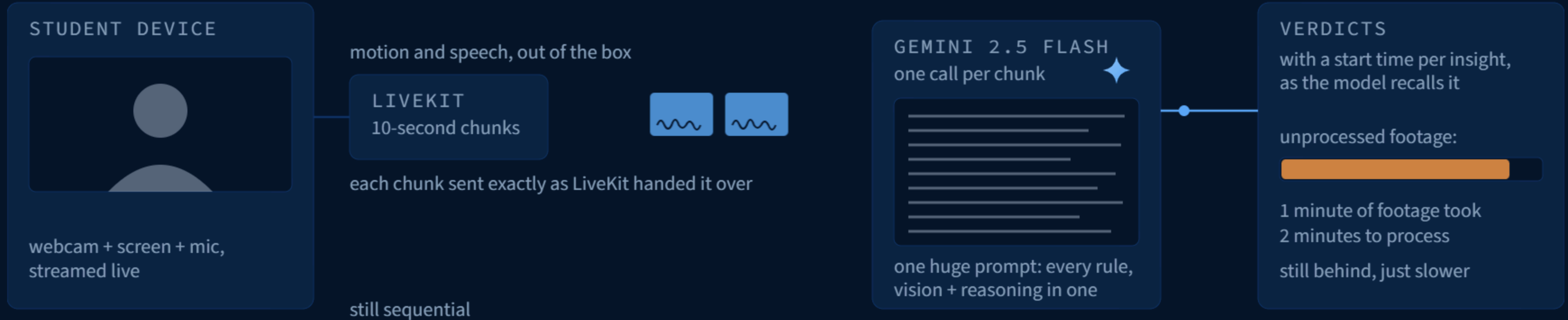
recall: stills are blind to motion and deaf to audio.

5x

latency: a backlog that compounds all session.

Blind to motion, deaf to audio, and \$1,000 per student per month.

Attempt 3: audio doubled recall and hid the precision gap



0.80

precision, on our annotated sessions. So we thought.

0.40

recall: audio and motion doubled it.

2x

latency: still compounding.

We shipped it everywhere, and our own number was wrong by a factor of eight.

In production July 2025 to August 2026: 72,000 hours of footage, on a few hundred students for most of that span.

“They are drinking from the phone”



PROCTORING INSIGHT

type: UnauthorizedDeviceUse
confidence: **HIGH**
duration: 8 sec

*“Student is holding a phone in their hand and appears to be **drinking from it** while the assessment is active.”*

90%

of flags reaching human review were false.

8

sessions in five weeks, same behavior, same flag.

HIGH

confidence, on every one of them.

No human would ever say a student is drinking from their phone.

Production human-review queue, June 2026: eight sessions, 8 April to 11 May 2026, each one a water bottle flagged as a phone. Reproduced in isolation on a three-line prompt. Photo: Markus, re-enacting it.

Prompting harder made it worse

FALSE ALARMS ON THE HARDEST SESSIONS, BY QUESTION SHAPE

“Is this person holding a phone?”

41% false

the production prompt: 400 lines of rules and exclusions

33%

“What object, if any, is in each hand?”

27%

describe first, classify the text after

27%

two inputs: the picture-in-picture recording, where the webcam is ~10% of the pixels, or the webcam stream alone · same model, same prompt: 67% → 40%

PRECISION, PROMPT VERSION BY PROMPT VERSION



Every prompt line fitted the prompt harder to the videos we had handpicked.

Measured on our hardest water-bottle sessions, May 2026; Gemini 3 models on the webcam-only input: 0 of 30 false. See Vo et al., “Vision Language Models are Biased”, ICLR 2026.

Making the pipeline dumber is what raised precision

CUT THE SESSION INTO WINDOWS. ASK EACH ONE QUESTION AT A TIME.



THE VISION PIPELINE, KEPT DUMB

one window, one rule, yes or no

stateless, parallel, replayable

one question per call

DETERMINISTIC CODE

When did the test start and end?

Is a calculator allowed right now?

no model at all

ONE SMART MODEL, AT THE END

reads all the text, builds the story of the test

reads text only, never sees a frame

after the test, not during

Separating the eyes from the judge cleared the gate

60 s window



Narrator

a person seated at a desk facing the screen

the screen shows a multiple-choice question

a dark rectangular handheld object raised to face height, screen side toward the person

no second person, no audible speech

Classifier

device in use? **yes**

someone helping? **no**

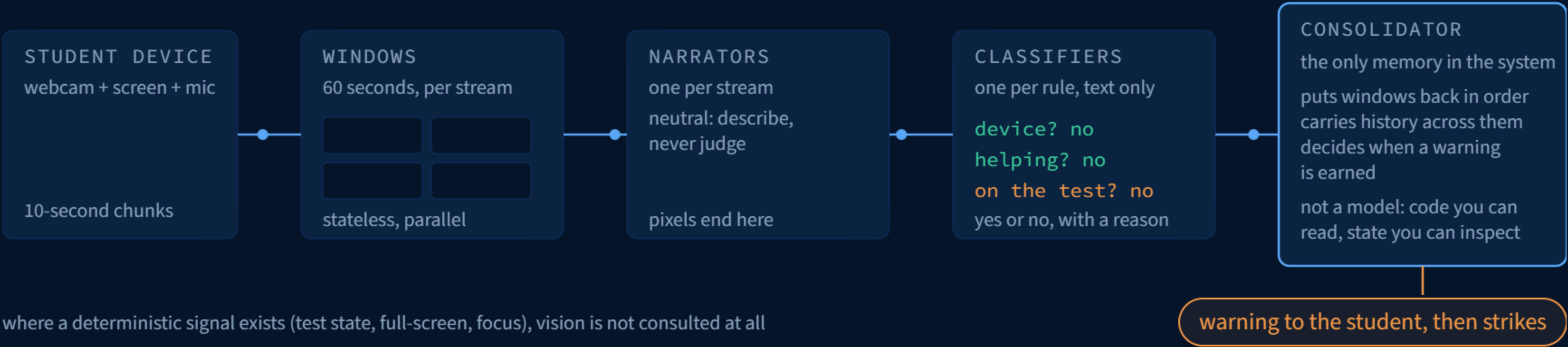
only the test on screen? **yes**

face in view? **yes**

The model that decides never sees a pixel.

Khayatan et al. 2026, When Prompts Override Vision: a question that presupposes an object is the largest single cause of hallucination, and describe-then-classify is the structural fix. We had shipped the same split before we read it.

The window graph holds 0.95 precision at a constant two minutes



0.95

precision on live traffic, against gold and human review.

0.90

recall, measured the same way.

2 min

from the event to the warning. Constant.

Stateless everywhere, except one place built to remember.

In production since the August 2026 cutover: 145,000 hours of footage. A detector reaches this rung by clearing 0.90 on the gold set, then 0.90 precision across 200 human reviews.

62 models auditioned for one job

THE FUNNEL, RE-RUN EVERY MONDAY

62 models watched across every vendor catalog

can watch a full minute of video

most cap at 5 images

allowed to see student video

US-hosted, DPA

baked against prod

same run, same sessions, 3 reps

above the floor

precision 0.90, at a carryable cost

1 job

THE CHAMPION • gemini-3.1-flash-lite

nobody beats it on precision, recall, cost and latency together

HOW AUDITIONS END

glm-5.3-flash **extremely high error rate**

pegasus-1.5 **quotas don't even support our tests**

nova-2-lite **never returns the schema it was asked for**

nova-pro **fires on half the hard tail**

gemma-4 **6.5 minutes to watch 1 minute**

claude-sonnet-5 **57x the cost**

gemini-3.8-flash **ties on quality, 2.4x the cost**

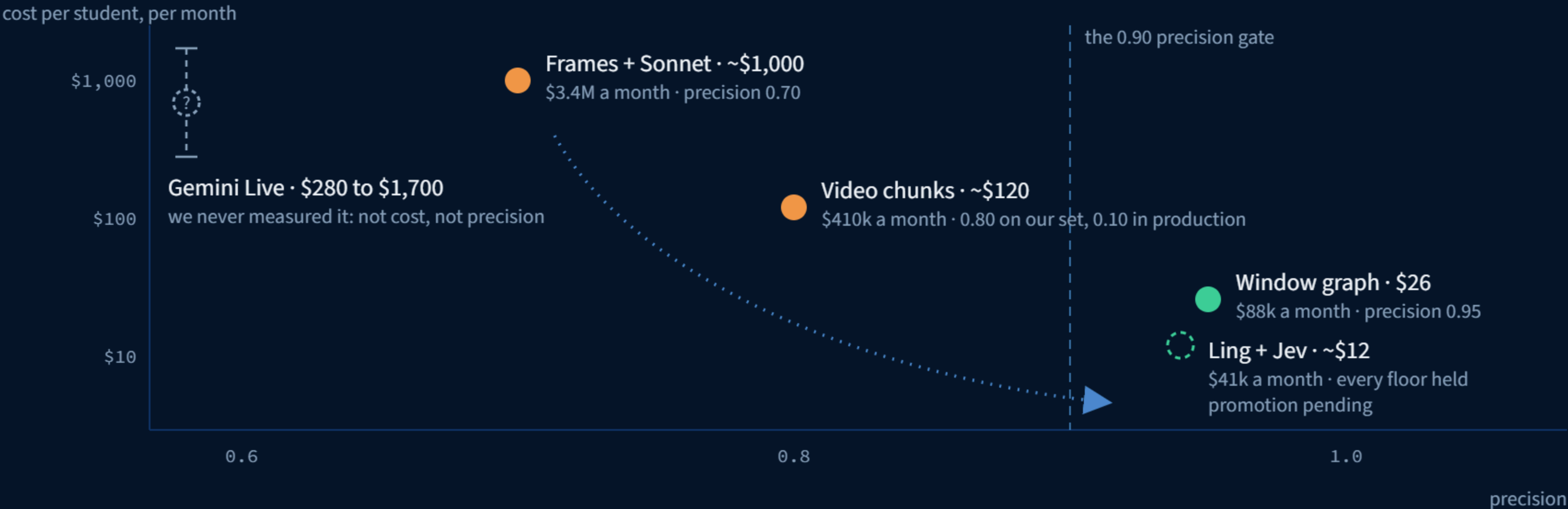
THE CHALLENGER, CONFIRMED THIS WEEK • Ling + TypeSafe Jev

holds every precision floor at half the cost; the promotion call is pending

Every new model gets a blind check first: what do you see? what do you hear?

The standing challenge: catch two more true windows than production on one insight, hold precision at 0.90, at a cost we can carry.

Cheaper and better are the same direction



The cheapest system we ever ran is the most accurate one. That is not a coincidence.

Vision model only, list prices, against a fixed unit: the normalized student at 34 recorded hours per month. Monthly totals at today's scale.

Six rules we would follow if we started again

1. Build gold before you build anything: annotate to align, and let the disagreements write the rules. BIAS 1
2. Set the bar before the architecture: precision is a gate, recall a target, latency a constant.
3. Keep it simple: no silver bullets, however strong the launch benchmarks looked. BIAS 2
4. The question shapes the answer: keep rule words away from the eyes, and your own bias out of the dataset. BIAS 5
5. Make the pipeline dumber: deterministic where code can decide, one binary question where it cannot. BIAS 4
6. Measure the input before you judge the model: check what it actually receives, and what it perceives. BIAS 3 BIAS 6

Every mistake was a bias we put there ourselves. That is the good news: so were all the fixes.



Thank you

Questions?

Serban Petrescu • Trilogy / TimeBack • spetrescu.ro